

Dr Artur Czeszumski

Data Scientist | Cognitive Neuroscientist | Lecturer



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Data scientist and cognitive neuroscientist with dual PhDs in Cognitive Science and Psychology. Works across football analytics, behavioral data, neurophysiology, and higher education. Combines rigorous experimental methods, statistical modelling, and data infrastructure to support decisions in both academic and applied settings.

Predictive modelling

Cognitive profiling

Behavioral and neurophysiological data

Experimental design

Data analysis and infrastructure

Experience

Jul 2026 - Present Product Analyst

Allegro

Warsaw, Poland • Product & Data

- Delivering analysis for the A/B testing platform, from experiment design to results interpretation.
- Ensuring statistical safeguarding of experiments, including power analysis and guarding against false positives.

A/B Testing

Statistical Safeguarding

SQL

Cloud Computing

AI

Experimentation

Data Analysis

Feb 2025 - Present Data and Cognition Scientist

Legia Warszawa

Warsaw, Poland • Research & Development

- Building and maintaining data infrastructure for academy performance workflows.
- Developing predictive models and cognitive profiling tools for player development.
- Supporting coaches and decision-makers with structured data pipelines and analysis.

Cognitive profiling

Cognitive tests

Data Analysis

Data infrastructure

Predictive modelling

Psychological tests

Football analytics

Mar 2026 - Present
Affiliated Resaercher

Maastricht, The Netherlands • Research



- Analysing EEG Hyperscanning Data

EEG

Preprocessing

Hyperscanning

Sep 2025 - Present
Lecturer

University of Warsaw
Warsaw, Poland • Education



- Teaching statistics, programming (Python), and data analysis.

Teaching

Programming

Python

Statistics

Data Analysis

Jan 2024 - Sep 2024
Postdoctoral Researcher

University of Bielefeld
Bielefeld, Germany • Research



- Contributed to developing an open database and live dashboard for behavioral and neurophysiological datasets.
- Advanced large-scale statistical modelling, visualization, and living analysis workflows.

Databases

Dashboards

Statistical modelling

Data visualization

Sep 2023 - Feb 2024
Football Data Analyst Intern

Double Pass
Mechelen, Belgium • Sport analytics



- Provided data-driven decision support for football development programs.
- Designed masterclasses on the use of data in the football industry.

Football development

Decision support

Workshops

Industry translation

Jan 2018 - Dec 2023
Joint PhD Researcher

Vrije Universiteit Amsterdam and University of Osnabrück
Amsterdam, Netherlands / Osnabrück, Germany • Research



- Researched cooperative behavior and neural markers of social interaction.
- Published peer-reviewed work in cognitive neuroscience, psychology, and human-machine interaction.
- Presented internationally, taught more than fifteen courses, and supervised more than thirty students.

EEG

Eye-tracking

Hyperscanning

Meta-analysis

Writing

Project Management

Data Analysis

Feb 2022 - Jun 2022

Researcher

Center for Brain, Mind and KANSEI Sciences Research

Tokyo, Japan • Research visit



- Developed networking solutions for EEG hyperscanning over cloud-based setups.

EEG

Networking

Hyperscanning

International collaboration

Dec 2021 - Jan 2022

Research Visit

Ngogo Field Station

Ngogo, Uganda • Field research



- Studied research practice on wild chimpanzee social interaction in a field context.

Field research

Social interaction

Comparative cognition

Dec 2019 - Feb 2020

Internship

Federal University of Rio Grande do Norte, Brain Institute

Natal, Brazil • Internship



- Analysed EEG time-series data in collaboration with local research teams.

EEG

Time-series analysis

Signal processing

May 2019

Research Visit

Perception and Learning Lab, KTH Stockholm

Stockholm, Sweden • Research visit



- Conducted a study on cooperative human-robot interaction.

Human-robot interaction

Cooperative behavior

Jan 2019 - Dec 2023

Associated Researcher

Situated Cognition Research Training Group

Osnabrück / Bochum, Germany • Research network



- Collaborated in an interdisciplinary research training environment on situated cognition.

Situated cognition

Interdisciplinary research

Aug 2014 - Dec 2017

Student Research Assistant

NeuroBioPsychology, University of Osnabrück

Osnabrück, Germany • Research support



- Managed lab operations, participant recruitment, and multimodal data acquisition.
- Preprocessed and analysed EEG, eye-tracking, and behavioral data.

Lab operations

EEG

Eye tracking

R / MATLAB / Python

Nov 2015 - Feb 2016

Student Research Assistant

Computational Linguistics, University of Osnabrück

Osnabrück, Germany • Research support



- Programmed survey and experiment workflows in LimeSurvey and analysed results in R.

Experiments

R

Survey programming

Oct 2013 - Dec 2013

IT Assistant

United Nations COP 19

Warsaw, Poland • Operations



- Supported hardware, software, and communications troubleshooting during the conference.

IT support

Troubleshooting

Event operations

Education

Jan 2018 - Dec 2023

Joint PhD in Cognitive Science and Psychology

University of Osnabrück and Vrije Universiteit Amsterdam

Osnabrück, Germany / Amsterdam, Netherlands



- Thesis: Neural foundations of cooperative social interactions.
- Research focused on social interaction, hyper-scanning, embodied cognition, and cooperative decision making.

Social neuroscience

Cooperative behavior

Hyperscanning

Machine Learning, AI

Apr 2014 - Aug 2017

MSc in Cognitive Science

University of Osnabrück

Osnabrück, Germany



- Advanced training in cognitive science, data analysis, statistics, and experimental methods.

Cognitive science

Data analysis

Experimental methods

Oct 2010 - Jan 2014

BSc in Cognitive Science

Nicolaus Copernicus University in Toruń

Toruń, Poland



- Undergraduate foundation in cognition, behavior, and interdisciplinary research methods.

Cognition

Behavioral science

Interdisciplinary methods

Publications

Peer-reviewed articles

Huang, A, M Pink, V Zemliak, **A Czeszumski**, and P König (2025). Choice history biases in dyadic decision making. In: *Scientific Reports* 15.1, p. 11420.

Kelsen, B, **A Czeszumski**, SHY Liang, YC Pei, J Hung, HL Chan, and HW Yeh (2025). Exploring foreign language anxiety and resting-state EEG alpha asymmetry. In: *Brain and Language* 261, p. 105519.

- Czeszumski, A**, SHY Liang, S Dikker, P König, CP Lee, SL Koole, and B Kelsen (2022). Cooperative behavior evokes inter-brain synchrony in the prefrontal and temporoparietal cortex: A systematic review and meta-analysis of fNIRS hyperscanning studies. In: *eNeuro* 9.2.
- Derakhshan, S, FN Nezami, MA Wächter, **A Czeszumski**, A Keshava, H Lukanov, MV De Palol, G Pipa, and P König (2022). Talking cars, doubtful users—A population study in virtual reality. In: *IEEE Transactions on Human-Machine Systems* 52.4, pp. 602–612.
- Solzbacher, J, **A Czeszumski**, S Walter, and P König (2022). Evidence for the embodiment of the automatic approach bias. In: *Frontiers in Psychology*, p. 4562.
- Walter, JL, JA Zerbe, V Schmidt, L Tiemann, M Mildt, F Pätzold, FN Nezami, P König, and **A Czeszumski** (2022). LightNet: a networking library to perform multi-participant experiments in Virtual Reality. In.
- Czeszumski, A**, F Albers, S Walter, and P König (2021). Let me make you happy, and I'll tell you how you look around: Using an approach-avoidance task as an embodied emotion prime in a free-viewing task. In: *Frontiers in psychology* 12, p. 604393.
- Czeszumski, A**, AL Gert, A Keshava, A Ghadirzadeh, T Kalthoff, BV Ehinger, M Tiessen, M Björkman, D Kragic, and P König (2021). Coordinating with a robot partner affects neural processing related to action monitoring. In: *Frontiers in Neurorobotics* 15, p. 686010.
- Pavlov, YG, N Adamian, S Appelhoff, M Arvaneh, CS Benwell, C Beste, AR Bland, DE Bradford, F Bublatzky, NA Busch, et al. (2021). # EEGManyLabs: Investigating the replicability of influential EEG experiments. In: *cortex* 144, pp. 213–229.
- Czeszumski, A**, S Eustergerling, A Lang, D Menrath, M Gerstenberger, S Schuberth, F Schreiber, ZZ Rendon, and P König (2020). Hyperscanning: a valid method to study neural inter-brain underpinnings of social interaction. In: *Frontiers in Human Neuroscience* 14, p. 39.
- Wahn, B, **A Czeszumski**, M Labusch, A Kingstone, and P König (2020). Dyadic and triadic search: Benefits, costs, and predictors of group performance. In: *Attention, Perception, & Psychophysics* 82.5, pp. 2415–2433.
- Czeszumski, A**, BV Ehinger, B Wahn, and P König (2019). The social situation affects how we process feedback about our actions. In: *Frontiers in Psychology* 10, p. 361.
- Wahn, B, **A Czeszumski**, and P König (2018). Performance similarities predict collective benefits in dyadic and triadic joint visual search. In: *PloS one* 13.1, e0191179.

Public Talks

Oct 2025

Conference talk

Artificial Intelligence (AI) Ally or enemy of entrepreneurship?

Print Unites | Uzhhorod, Ukraine

Nov 2024

Conference talk

Cognitive neuroscience beyond talent development

Science4Football | Warsaw, Poland

Jul 2022

Conference talk

2nd Person Approach To Studying Cooperative Social Interactions

Interpersonal Neuroscience: Connecting People via Their Bodies and Brains | Online

Jul 2022

Conference talk

2nd Person Approach To Studying Cooperative Social Interactions

ESCAN 2022 | Vienna, Austria

Jul 2021	EEG hyperscanning workshop
Summer school lecture	Interpersonal Neuroscience: Connecting People via Their Bodies and Brains Online
Jun 2021	Coordinating with a robot partner affects action monitoring related neural processing
Conference talk	Virtual Social Interaction Conference Online
Oct 2020	Hyperscanning: A valid method to study neural inter-brain underpinnings of social interaction
Conference talk	Neuromatch Online
Jun 2020	Neural underpinnings of social interactions
Lab seminar	NeuroPsychoLinguistics Lab Seminar Geneva, Switzerland
Jun 2020	Neural underpinnings of social interactions
Lab seminar	Emotion Regulation Lab Seminar Amsterdam, Netherlands
Mar 2019	The social situation affects how we process feedback about our actions
Conference talk	Understanding Others: Technical Issues Related to Real-Time Social Interaction Phenomena Munich, Germany
Feb 2019	The social situation affects how we process feedback about our actions
Conference talk	Cognitive Science Arena Brixen, Italy Awarded 2nd prize for best talk.
Sep 2018	The social situation affects how we process feedback about our actions
Conference talk	Social Cognition in Humans and Robots Hamburg, Germany
Oct 2017	Does the social situation affect how we process feedback about our actions?
Conference talk	Understanding Social Cognition 2017 Lublin, Poland

Additionally, I have presented more than 30 posters at scientific meetings.

Teaching Experience

I have supervised more than 30 BSc and MSc theses.

2026	Data Analysis (Python) University of Warsaw BSc
2025	Python Foundation University of Warsaw BSc
2025	Python Project University of Warsaw BSc
2021/22	Study project: Dyadic Decision Making (DDM) University of Osnabrück MSc
2021/22	Social Interactions in VR University of Osnabrück MSc
2021/22	Study Project: Dyadic Decision Making (Part II) University of Osnabrück MSc
2021	Interpersonal Neuroscience: Connecting People via Their Bodies and Brains - Hyperscanning EEG Workshop Vrije Universiteit Amsterdam Summer School / Workshop
2020/21	Begleitseminar zum Studienprojekt: Hyperscanning University of Osnabrück MSc
2020/21	Dyadic Interactions in VR University of Osnabrück MSc
2020/21	Independent Study Course University of Osnabrück MSc
2020/21	Study Project: Hyperscanning (Part II) University of Osnabrück MSc

2019/20	Study Project: Hyperscanning (Part I) University of Osnabrück MSc
2019/20	Interdisciplinary Course: Hyperscanning University of Osnabrück MSc
2018	Advanced Topics in Action and Cognition University of Osnabrück MSc
2018	Statistical Rethinking Reading Club University of Osnabrück MSc

Awards

March 2021	Award for Best Poster Presentation Mind Brain Body Symposium Berlin, Germany Hyperscanning: A Valid Method to Study Neural Underpinnings of Social Interaction
July 2020	FENS 2020 Accommodation and Travel Grant FENS Forum Grant originally awarded for travel and accommodation; converted into a waived conference fee due to the COVID-19 pandemic. Associated work: Hyperscanning: A Valid Method to Study Neural Underpinnings of Social Interaction.
February 2019	Second Prize for Best Talk Cognitive Science Arena Brixen, Italy The Social Situation Affects How We Process Feedback About Our Actions
October 2018	Gentner Symposium Travel and Accommodation Grant Gentner Symposium - Understanding Others: From Psychological Concepts to Neural Mechanisms Jerusalem, Israel Dynamics of Feedback Processing in Cooperative and Competitive Situations